TEJASVI VASANT HEGDE

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PROFESSIONAL SUMMARY

Computer Science student with a strong foundation in software engineering and machine learning with hands-on exposure to building AI-powered, real-time and full-stack applications. Proficient in Python, TensorFlow, FastAPI, React.js and Docker with a focus on developing practical and scalable software solutions.

EDUCATION

Sep 2023 - Present

B.E. in Computer Science | **R. V. College of Engineering, Bengaluru**
CGPA: 8.91/10

2021 - 2023

Pre-University (PCMC) | **Ambika PU College, Puttur**
Percentage: 96.17% , KCET - 156

PROFESSIONAL EXPERIENCE

Mar 2026 – Jul 2026

Intern - (CPP) | **Hewlett Packard Enterprise (HPE)**
Remote | Docker, Redis, RAG, LangGraph, Apache Airflow, Bash

- Developed an AI-powered deployment platform using Airflow, LangGraph and RAG for automated deployments and failure analysis.
- Built a self-healing pipeline using ChromaDB and AI agents for automated DAG patching and SSH-based remediation.
- Developed a React and FastAPI dashboard for deployment management, live log monitoring, and AI-powered operational insights.

PROJECTS

Mar 2026 – Jun 2026

Electricity Load Forecasting | GitHub
Python, React.js, LightGBM, TensorFlow (GRU), FastAPI, Pandas, Tailwind CSS

- Developed a hybrid GRU-LightGBM stacking model for short-term electricity load forecasting using 17,376 hourly observations from Karnataka SLDC along with weather, holiday, and temporal features to predict demand up to 7 days ahead.
- Engineered 62 features and achieved 5.59% MAPE, reducing error by 29% over the standalone GRU baseline.
- Developed an end-to-end FastAPI + React forecasting platform with forecast analytics, feature-importance analysis, and an LLM-powered What-If simulation engine for evaluating demand scenarios.

Mar 2026 – Jun 2026

Remote Voting System | GitHub
React.js, Node.js, Express.js, MySQL, Raspberry Pi, Python, JWT, AES-256, Face Recognition

- Developed a location-independent secure voting system with facial authentication and tamper-resistant vote storage.
- Built the React, Node.js, Express and MySQL stack with JWT authentication, AES-256 encryption, and real-time administration APIs.
- Integrated a Raspberry Pi terminal with camera-based face verification, GPIO controls and an OLED display.

Oct 2025 – Jan 2026

Deep Learning-Based Image Steganography | GitHub
Python, PyTorch, CNNs, GANs, U-Net, Flask, React.js, TypeScript, SSIM

- Developed an adversarial Encoder-Decoder-Discriminator architecture with U-Net style skip connections, residual blocks, and SE attention to hide secret images within cover images while preserving visual imperceptibility.
- Combined L1, MSE, SSIM, perceptual, and adversarial losses, achieving 35.71 dB PSNR and 0.8845 SSIM on cover to stego images.
- Conducted ablation, payload-stress and robustness experiments across discriminator and loss components, embedding strengths and JPEG compression and implemented the model using a Flask and React.js web application for real-time image embedding and extraction.

TECHNICAL SKILLS

Languages: C, Python, JavaScript, HTML, CSS
Frameworks: React, Node.js, FastAPI
Machine Learning: LightGBM, TensorFlow, Pandas, RAG
Tools: Docker, Redis, Airflow, Bash, Git
Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks

CERTIFICATIONS & ACHIEVEMENTS

- **3rd Place, D3 Hackathon (State Level)** — IIIT Bhubaneswar
- **AI for Economics** — NPTEL
- **Data Science for Engineering** — NPTEL
- **Complete Full Stack Web Development** — Udemy